

Eventual Norm Continuity Does Not Preserve the Spectral-Gap Necessity for Eventual Domination

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Status. Candidate counterexample, likely valid. This gives a full negative answer to the first explicit question in source Remark 3.3: implication (i) $\Rightarrow$ (iii) of Theorem 3.1 fails when analyticity is replaced by eventual norm continuity. The separate question in the same remark about removing one smoothing assumption is not resolved here.

1 Source question and answer

Glück and Mugnolo characterize individual eventual domination between two distinct real analytic semigroups under spectral, smoothing, and eventual positivity hypotheses [1, Theorem 3.1]. Their assertion (i) says that for every $0 < f$ the orbit of B eventually dominates the nonnegative orbit of A ; assertion (iii) is the strict spectral gap $s(B) > s(A)$. Immediately after the proof they ask whether (i) $\Rightarrow$ (iii) remains true when analyticity is weakened, for instance to eventual norm continuity.

from E to E_u due to the closed graph theorem. Hence,

$$\|e^{tA}\|_{E \rightarrow E_u} \leq \|e^{t_0 A}\|_{E \rightarrow E_u} \|e^{(t-t_0)A}\|_{E \rightarrow E} \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

Therefore, there exists a time $t_3 \in [0, \infty)$ such that $\|e^{tA}f\|_u \leq \frac{\varepsilon}{4}$ for all $t \geq t_3$, and thus we have $|e^{tA}f| \leq \frac{\varepsilon}{4}u$ for all $t \geq t_3$. Using that the semigroup $(e^{tA})_{t \in [0, \infty)}$ is real, we thus conclude that $e^{tA}f \leq \frac{\varepsilon}{4}u$ for all $t \geq t_3$.

Consequently, we have $e^{tB}f - e^{tA}f \geq \frac{\varepsilon}{2}u - \frac{\varepsilon}{4}u \gg_u 0$ for all $t \geq t_1 := \max\{t_2, t_3\}$. This proves assertion (ii).

“(ii) $\Rightarrow$ (i)” This implication is obvious. $\square$

Remark 3.3. In the proof of Theorem 3.1, the analyticity assumption on the semigroups is only needed in its full strength to employ the identity theorem for analytic mappings at the end of the proof of implication “(i) $\Rightarrow$ (iii)”. We do not know whether this implication remains true if the analyticity assumption is weakened (for instance, to eventual norm continuity).

The implication “(iii) $\Rightarrow$ (ii)” remains true if we only assume that both semigroups $(e^{tA})_{t \geq 0}$ and $(e^{tB})_{t \geq 0}$ are eventually norm continuous rather than analytic.

The assumptions $e^{t_0 A}E \subseteq E_u$ and $e^{t_0 B}E \subseteq E_u$ in Theorem 3.1 can be omitted for the implication “(i) $\Rightarrow$ (iii)”. We do not know whether the assumption $e^{t_0 B}E \subseteq E_u$ is really needed for the converse implication “(iii) $\Rightarrow$ (ii)” to hold; but the following example shows that at least the assumption $e^{t_0 A}E \subseteq E_u$ cannot be dropped, in general.

Figure 1: Source PDF page 9, Remark 3.3. The first displayed open question is answered negatively below.

The answer is no. The mechanism is finite-time erasure: two different semigroups may have exactly the same tail without being the same semigroup. We retain every other hypothesis of the source theorem, including nonempty spectrum, pole conditions, both smoothing inclusions, and the stated positivity assumptions.

2 A nilpotent shift with empty generator spectrum

Let $H = L^2(0, 1; \mathbb{C})$ and define the left-shift semigroup $(S(t))_{t \geq 0}$ by

$$(S(t)g)(x) = \begin{cases} g(x+t), & x+t \leq 1, \\ 0, & x+t > 1. \end{cases}$$

It is a real C_0 -semigroup and $S(t) = 0$ for $t \geq 1$. Its generator is

$$Gg = g', \quad D(G) = \{g \in H^1(0, 1) : g(1) = 0\}.$$

Lemma 1. *The spectrum of G is empty.*

Proof. For every $\lambda \in \mathbb{C}$ and $h \in H$, set

$$(K_\lambda h)(x) = \int_x^1 e^{\lambda(x-s)} h(s) ds.$$

This bounded Volterra operator maps H into $D(G)$. Differentiation gives $(K_\lambda h)' = \lambda K_\lambda h - h$, so $(\lambda - G)K_\lambda h = h$. Conversely, integration of the scalar differential equation with terminal value $g(1) = 0$ gives $K_\lambda(\lambda - G)g = g$ for $g \in D(G)$. Hence $K_\lambda = \mathcal{R}(\lambda, G)$ for every complex λ , proving $\sigma(G) = \emptyset$. $\square$

3 Counterexample

On the Banach lattice $E = L^2(0, 1; \mathbb{C})$, put $u = \mathbf{1}$ and define

$$\varphi(f) = \int_0^1 f(s) ds, \quad Pf = \varphi(f)\mathbf{1}, \quad Q = I - P, \quad F = \ker \varphi.$$

Thus P is a real rank-one projection, $E = \text{span}\{\mathbf{1}\} \oplus F$, and F is a real-invariant separable infinite-dimensional Hilbert space. Choose a complex-linear unitary $J : H \rightarrow F$ which maps real functions to real functions; such a J is obtained by choosing real orthonormal bases and complexifying. Write

$$W(t) = JS(t)J^{-1} \quad \text{on } F.$$

Finally, define two operator families on E by

$$T_A(t) = P + W(t)Q, \quad T_B(t) = P + W(2t)Q. \quad (1)$$

Theorem 1. *The families in (1) are distinct real C_0 -semigroups that satisfy every assumption of source Theorem 3.1 after “analytic” is replaced by “eventually norm continuous.” Assertion (i) of that theorem holds, but*

$$s(A) = s(B) = 0,$$

so assertion (iii) fails.

Proof. Semigroup law, reality, and distinctness. Relative to $E = \text{span}\{\mathbf{1}\} \oplus F$, the two families are unitarily similar to

$$1 \oplus S(t) \quad \text{and} \quad 1 \oplus S(2t),$$

respectively. They are therefore real C_0 -semigroups. They are distinct: if $g = \mathbf{1} \in H$ and $h = Jg \in F$, then at $t = \frac{1}{4}$ their complementary components are $J\mathbf{1}_{(0, 3/4)}$ and $J\mathbf{1}_{(0, 1/2)}$.

Eventual norm continuity and smoothing. Nilpotence of S gives

$$T_A(t) = P \quad (t \geq 1), \quad T_B(t) = P \quad (t \geq \tfrac{1}{2}).$$

Both semigroups are thus constant in operator norm for all sufficiently large times. Moreover

$$E_u = L^\infty(0, 1), \quad T_A(1)E = T_B(1)E = PE = \text{span}\{\mathbf{1}\} \subseteq E_u.$$

This verifies both smoothing inclusions with the common time $t_0 = 1$.

Spectra and peripheral poles. The generators are, under the same decomposition,

$$A = 0 \oplus JGJ^{-1}, \quad B = 0 \oplus 2JGJ^{-1}.$$

Lemma 1 and its scaled version yield $\sigma(A) = \sigma(B) = \{0\}$. More precisely, near zero,

$$\begin{aligned} \mathcal{R}(\lambda, A) &= \lambda^{-1}P + J\mathcal{R}(\lambda, G)J^{-1}Q, \\ \mathcal{R}(\lambda, B) &= \lambda^{-1}P + J\mathcal{R}(\lambda, 2G)J^{-1}Q. \end{aligned}$$

The second terms are operator-valued entire functions of λ . Thus 0 is a simple pole for both generators, their peripheral spectra are $\{0\}$, and both spectral bounds equal zero. In particular, the construction does not evade the source's nonempty-spectrum condition.

Positivity and domination. If $0 < f \in E$, then strict positivity of integration gives

$$Pf = \varphi(f)u \gg_u 0.$$

Hence $(T_A(t))$ is uniformly eventually positive and $(T_B(t))$ is uniformly eventually strongly positive with respect to u ; these are stronger than the two source positivity assumptions. Finally, for every $0 < f \in E$ and every $t \geq 1$,

$$T_B(t)f = T_A(t)f = Pf \geq 0.$$

Source assertion (i) therefore holds, even with a time independent of f , while the strict inequality in (iii) fails. This completes the hypothesis crosswalk and the counterexample. $\square$

4 Why analyticity is the exact fault line

In the equal-spectral-bound case, the published proof eventually derives $T_B(t)f = T_A(t)f$ for all sufficiently large t and every $0 < f$. Analytic unique continuation then extends this equality to every positive time, contradicting distinctness. Eventual norm continuity has no analogous backward-uniqueness property. In the example above the complementary subspace F is erased in finite time, after which two different histories are indistinguishable.

The semigroups above cannot be analytic. Indeed, $T_A(t) = P$ for $t \geq 1$ has the nontrivial kernel F , whereas every positive-time operator of an analytic C_0 -semigroup is injective: if $T(t_0)x = 0$, the semigroup law makes the analytic orbit vanish on a tail, the identity theorem makes it vanish on $(0, \infty)$, and strong continuity at zero gives $x = 0$.

5 Deep-upgrade attempt on the second source question

Remark 3.3 separately asks whether the inclusion $e^{t_0 B}E \subseteq E_u$ is needed for (iii) $\Rightarrow$ (ii). A later refinement by Arora and Glück proves domination without that B -smoothing inclusion if one directly assumes that for every $0 < f$ there are $c_f > 0$ and t_f such that

$$e^{tB}f \geq c_f u \quad (t \geq t_f)$$

[2, Proposition 4.2]. This is stronger than individual eventual strong positivity: in the latter definition the witness in $e^{tB}f \gg_u 0$ may depend on t .

The additional attack tested whether spectral convergence or a Baire argument forces such a time-uniform lower constant. Strong convergence in E does not control the gauge norm of E_u , whose positive order interior is generally not open in the ambient norm; this is precisely where the source proof uses B -smoothing. Conversely, a moving-spike construction must obey both the semigroup law and eventual strong positivity for every positive initial vector, and no rigorous example survived those constraints. Thus the second question remains outside the scope of the present full counterexample to the first question.

6 Novelty and verification

The proof was audited in nine separate checks: the nilpotent semigroup law, the all-parameter resolvent formula, the real unitary decomposition, the two semigroup laws, distinctness, spectrum and pole order, smoothing, positivity, and the domination quantifiers. No numerical approximation is used.

A bounded search on 11 August 2026 covered the run indexes, the local parsed full-source corpus, the exact source wording, the exact arXiv id and title, and close arXiv searches involving eventual norm continuity, eventual equality, nilpotent shift semigroups, empty spectrum, and rank-one tails. The later refinement [2] was also inspected directly. No prior answer to the first source question or matching construction was found. The construction is elementary, so novelty confidence is medium; mathematical validity confidence is high pending human review.

Human review focus. Check the real-unitary identification J , the resolvent formula in Lemma 1, the direct-sum pole calculation, and the exact match with every hypothesis of source Theorem 3.1 under the proposed regularity weakening.

References

- [1] J. Glück and D. Mugnolo, *Eventual domination for linear evolution equations*, Math. Z. 297 (2021), 301–330; [arXiv:1912.13057v2](#).
- [2] S. Arora and J. Glück, *Criteria for eventual domination of operator semigroups and resolvents*, Operator Theory: Advances and Applications 292 (2023), 1–26; [arXiv:2204.00146v3](#).